
Experiment No: 1

Cathode Ray Oscilloscope I

Aim

To familiarize with cathode ray oscilloscope and to study the functions.

Apparatus:

CRO, Function generator and pobs.

Theory

Cathode ray oscilloscope, CRO, is an instrument which gives the visual representation of electrical signals. It is one of the ways versatile tools used for research and study of electronic circuits and systems. It displays the signals on the screen in X axis and Y axis which is used in conventional graphs. X axis represents the time and Y axis represents the amplitude of the signal. Oscilloscopes are capable for displaying voltage variations which take place over a period of micro or nano seconds.

Cathode ray tube is the heart of a CRO. It is a vacuum tube which generates narrow electron beam which is made to fall on a fluorescent screen at one end of the tube. The electron beam is deflected horizontally and vertically by horizontal and vertical deflection plates respectively.

Power controls:

OFF/ INTENS: Turn power on and pilot LED indicates the status whether it is on or off. The intensity of the displayed electron beam can be varied using this knob.

FOCUS: Using this, beam can be made sharply defined. This helps to take the readings more accurately.

Vertical Deflection-

Y-POSITION: Control the vertical shifting of the trace

AMP/DIV: Enlarge the signal in vertical direction. Knob can be set in ranges from 2mv/div to 10v/div

ON/OFF: Turns the channel on or off.

AC/DC: In AC position Dc component of the signal blocked by a blocking capacitor. In DC position, signal is directly coupled.

Time Base and Horizontal Deflection

X-POSITION: Controls the horizontal shifting of the trace

LEVEL: Helps the signal to remain still

S/MS: in combination with TIME/DIV switch select the time coefficients.

MAGN $\times 10$: Allows the magnification of the horizontal deflection by a factor of n.

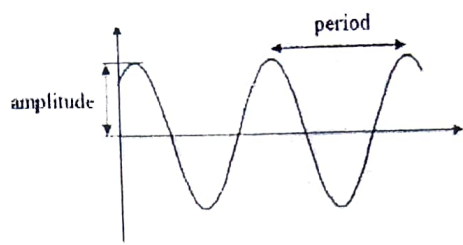
X A: Horizontal deflection is determined by the input signal to channel A

CAL Knob: Knob should be turned to CAL position to take readings.

INT/EXT: An external triggering signal for timer-base can be given through EXT socket.

NORM/TV: In NORM position, normal triggering is obtained. In TV position, TV line or TV frame synchronization is obtained.

Diagram



Observations

Measurement of Amplitude:-

Sl no:	Voltage	No: of divisions	Volt/ division	Amplitude
1				
2				
.				
.				
.				
.				

Measurement of Frequency:-

Sl no:	frequency	No: of divisions	Time/ division	Frequency
1				
2				
.				
.				
.				
.				

Procedure

To measure amplitude of signal

- Switch on the CRO, obtain a sharply defined trace of a horizontal line on the screen by adjusting INTENS and FOCUS knobs
- Adjust the Y-position knob so as to make coincide with the center line on the screen by keeping AC-DC switch in GND position.
- Connect the function generator with the CRO using a probe and switch on the function generator. [CRO probe is a coaxial chord with BNC (British Navel Connector) clip at one end and two crocodile clips at other end]
- Count the number of divisions occupied by the signal from peak to peak. Multiple these by the value indicated by the AMP/DIV knob. This gives the peak to peak amplitude of the signal.
- Repeat the above steps for various settings of AMP/DIV knob.

To measure frequency of signal

- Obtain a sharply defined trace of horizontal line on the screen by adjusting INTENS and FOCUS knobs. Feed the signal whose frequency is to be measured, to either of the channels using a probe and observe the signal on CRO.
- Adjust the TIME/DIV knob so as to see two or three cycle of the waveform. Count the number of divisions in one cycle of the waveform. Multiply this by the time/div setting. This is the time period of the signal. Its reciprocal gives frequency of signal.
- Repeat the above steps of various settings of TIME/DIV knob.

Result:

- Familiarize with cathode ray oscilloscope and study the functions.
- Amplitude of a give wave is measured and it is remains constant for a particular frequency
- Frequency of a given signal also measured.

Experiment No: 8

Fraunhofer Diffraction at Single Slit - I

Aim

To sketch the Fraunhofer diffraction pattern by detecting diffracted light using LDR and travelling microscope.

Apparatus Used

Laser 625nm, 0.5mWatt with micrometer adjustable slit, travelling microscope fitted adjustable stand and LDR, Digital micro-ammeter 0-200 μ A

Theory

Fraunhofer diffraction is a special case of Fresnel diffraction in which the source illuminating the aperture and the observation screen are located at infinite distance. The LASER provides coherent source of light. The screen is placed 1-2 meters to be used as observer, from the source. Intensity is recorded by a LDR placed behind the receiving slit. The LDR is energized by a regulated power supply and digital dc micro-ammeter is used to measure the LDR current that is proportional to the light intensity it receives.

The intensity of the diffracted light coming out of the emitter slit is given by

$$I = I_0 \left(\frac{\sin \beta}{\beta} \right)^2$$

$$\beta = \frac{\pi a}{\lambda} \sin \psi$$

ψ - Wave function

a - slit width

λ - Wave length

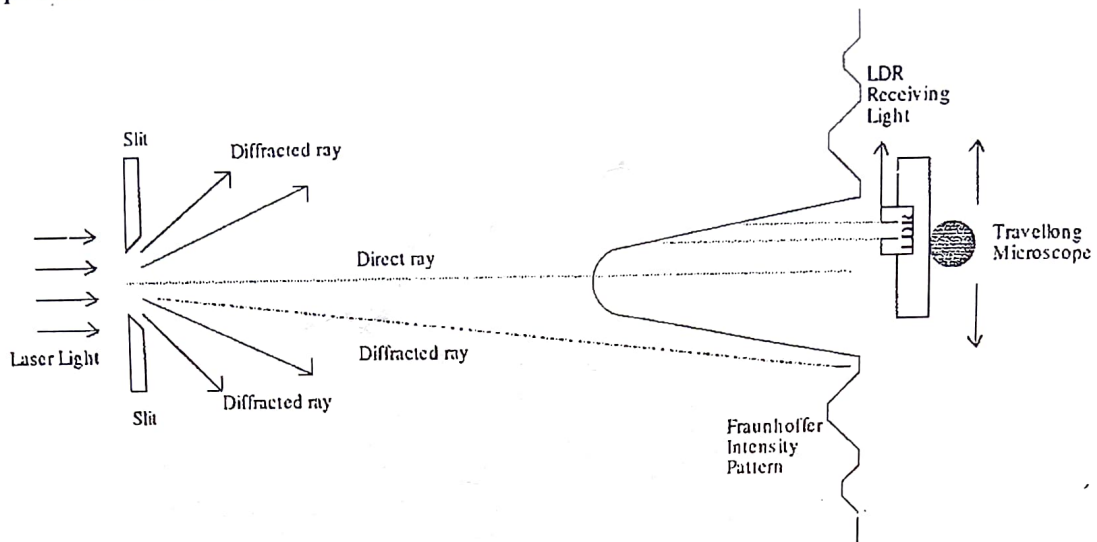
The diffraction pattern is a series of bright and dark spots. The position of the maxima or the brighter spot and the minima the dark spot is obtained by equating the derivatives of equation to zero

$$\sin \beta (\sin \beta - \beta \cos \beta) = 0$$

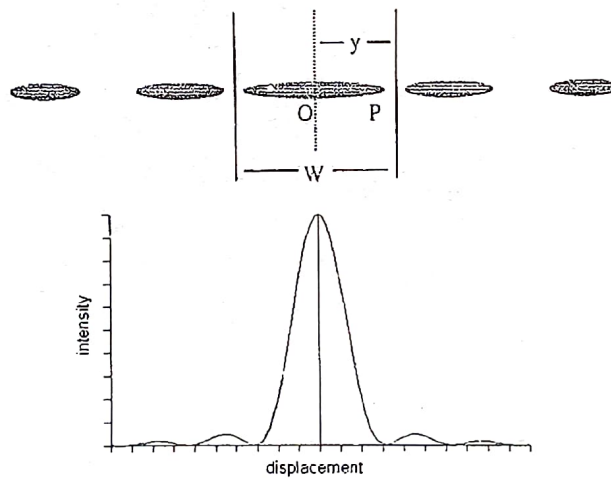
When $\sin \beta = 0$ maxima occurs, or when $\beta = 0, \pi, 2\pi, 3\pi, 4\pi$ etc.

Minima occurs at $(\sin \beta - \beta \cos \beta) = 0$ this equation doesn't have exact solution except for $\beta = 0$, the other approximate solutions are $\beta = 4.49, 7.73, 10.9$ radians etc

Experimental arrangement:-



Diffraction pattern:-



Observations:-

Least count of the micrometer = _____

Sl: no:	Distance (mm)			LDR current (μA)
	MSR	VSR	Total	
1				
.				
.				
.				

Procedure:

- The least count of the micrometer adjustable slit is determined following standard procedure.
- The micrometer adjustable slit is mounted on a stand and placed in front of the laser source
- The travelling microscope containing LDR detector, behind the receiver slit is placed on the way of the laser light and laser light is focused by adjusting stand on the receiver slit.
- The laser light is correctly focused to the receiving slit. Micro-ammeter and power supply are connected to LDR and current is noticed in the micro-ammeter.
- A white paper is now held in front of the receiving slit and diffraction pattern is observed. The micrometer source slit or the emitting slit is adjusted if need to get a faithful diffraction pattern.
- The reading of the micrometer source slit is noted and recorded. The Travelling microscope is moved to the left from about 3cm from center position where the current is maximum. At this position micro-ammeter reading and travelling microscope readings are noted and recorded.
- The travelling microscope is moved towards the right by 1mm and micro-ammeter reading is noted. This is continued till the maximum intensity is obtained and the corresponding readings obtained are recorded.
- After reaching the maximum intensity position, movement of the travelling microscope is further continued in steps of 1mm. This is done until second and third maxima are reached.
- The distance between the laser source and the travelling microscope is measured using a meter scale. This is the distance D
- A graph is drawn taking distance moved along X-axis and current along Y-axis

Result

Fraunhofer diffraction pattern is sketched in this experiment using LDR as light detector. The diffraction pattern obtained is the photocopy of the theoretical predictions.

Experiment No: 10

Numerical Aperture of Optic Fiber Cable ✓

Aim

To determine numerical aperture using IEEE 1394 fire wire cable (OFC) and 6.5 diode laser

Apparatus used

OFC 1.5m (IEEE 1394 fire wire cable), diode laser red 625nm, travelling microscope bed carrying fixed screen and a movable chuck.

Theory

Numerical aperture is one of the important parameter of a cable. In optics, aperture refers to opening of light through which light passes and falls on the optics. Hence aperture has dimensions of area [L^2]. Light from the ofc comes out of the cable and falls on the detector. As the name implies, NA is a number which varies between 0 and 1. In other words, the Na implies the amount of light linking the cable and the detector or the source. Smaller the NA better will be the cable. NA of the cable is defined as sine of the half angle of the cone generated due to the divergence of rays as shown.

$$NA = \sin \theta$$

Light coming out of OFC falls on the screen kept at a distance L from it. An image of the laser spot is seen. This spot and emerging light forms a cone. If D is the diameter of the circular spot, 'f' is the distance between screen and OFC, then

$$\tan \theta = \frac{D/2}{f} = \frac{D}{2f}$$

$$\theta = \tan^{-1} \left(\frac{D}{2f} \right)$$

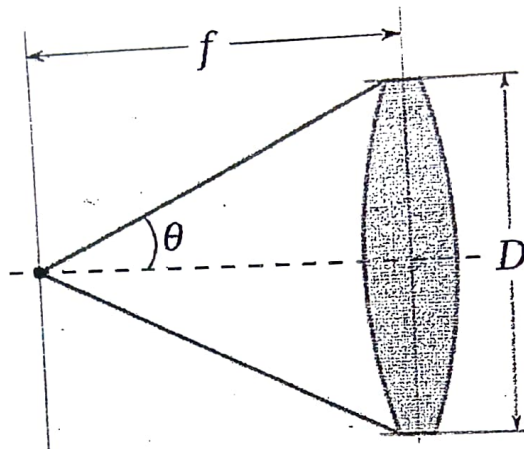
By measuring D and f, θ can be determined. Hence the NA can be calculated by the equation,

$$NA = \sin \theta = \sin \left[\tan^{-1} \left(\frac{D}{2f} \right) \right]$$

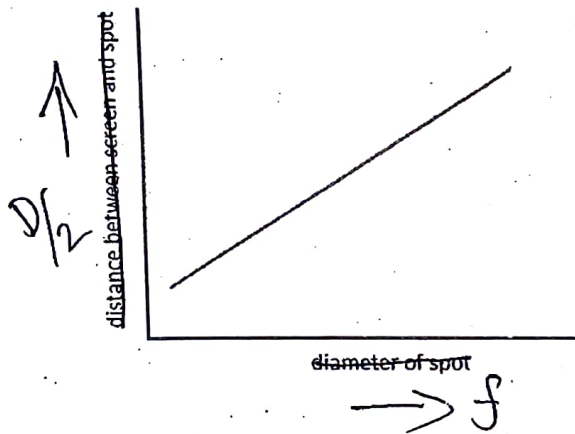
Procedure

1. The optical cable is connected to the laser and the laser light coming through other end of the cable verified.
2. In other end of the cable is coupled to the chuck fixed on the transverse motion bench.
3. The chuck carrying the OFC is brought close to the screen and the laser spot is seen on the screen as shown.
4. By adjusting the line screw of the microscopic bench the spot size is 8mm
Spot size = D = 8mm
5. The distance between the fixed screen and chuck carrying OFC is noted on the graduated scale fixed along the X axis.

Diagram



Model Graph



Observations

Cable	Sl. no:	D	$D/2$	f	$\tan \theta = \left(\frac{D}{2f}\right)$	θ	$\sin \theta$
1	1						
	2						
	.						
	.						
2	1						
	2						
	.						
	.						

6. The trial is repeated by increasing the spot size like 10mm, 12mm, 14mm... and the corresponding 'f' is noted. The readings obtained are tabulated.
7. The experiment is repeated with another cable of length 1m and the readings obtained are tabulated.
8. A graph is plotted showing the variation of D and f as shown.

Result

Numerical aperture of the fiber optic cable (), NA =

Numerical aperture of the fiber optic cable (), NA =

Numerical aperture obtained from graph for the cable () =

Numerical aperture obtained from graph for the cable () =

Experiment No: 5

Solar Cell IV Characteristics

Aim

To study the iv characteristics curves of monocrystalline silicon solar cell using standard 50W MR16 type Halogen lamp (standard line Dichroic)

Introduction

Solar cells are special semiconductor devices, which produce electric voltage across its terminals by absorbing sunlight. The light photons are converted in to voltage hence the process is called photovoltaic effect. Semiconductors such as n-Si, p-Si, n-GaAs, SnO₂, Au-GaAs, Au-GaP, Au-CdS, exhibit photovoltaic effect. A solar cell looks like a fragile wafer. Monocrystalline silicon is the popular commercial grade material used for the manufacture of solar cell. Solar cells are cascaded in series or parallel to obtain solar cell module for commercial use.

The photons absorbed by the solar cell create hole-electron pair inside the crystal. Accumulations of such hole-electron pair give rise to potential across the terminals of the solar cell. Figure-2 shows equivalent circuit of a solar cell. This is the simplified equivalent circuit generally used for circuit analysis. The solar cell equivalent circuit consists of constant current source producing photocurrent I_{ph} and diode parallel to it. The photo current I_{ph} is directly proportional to light energy falling on it.

Applying Kirchoffs law the various currents of the solar are related as

$$I_{ph} = I_D + I$$

Where I_{ph} is the photocurrent

I_D is the current through the diode

I is the output current

The output current I is given by

$$I = I_{ph} - I_D$$

The diode current at room temperature 300°K is given by

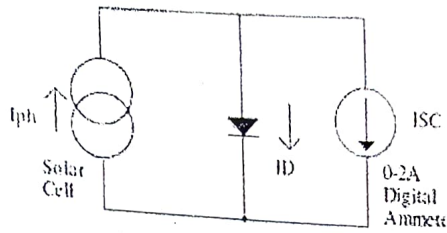
$$I_D = I_s (e^{40V} - 1)$$

Where I_s is the reverse saturation current and V is voltage across the output terminal.

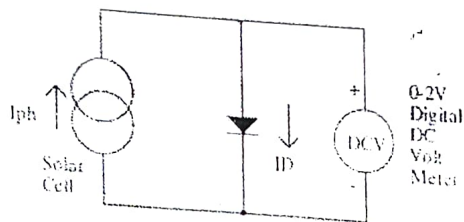
$$I = I_{ph} - I_s (e^{40V} - 1)$$

This Equation represents I-V characteristics of a solar cell.

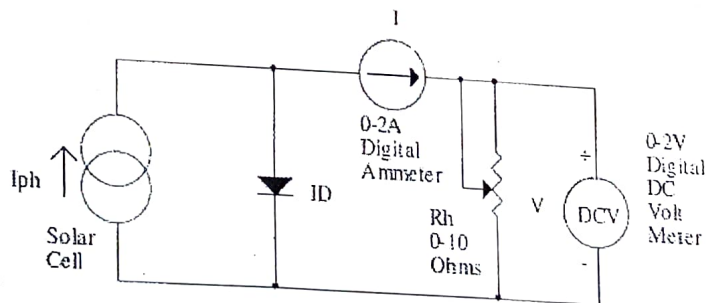
Circuit diagrams



Determination of I_{sc}



Determination of V_{oc}



Circuit Diagram I-V Characteristics

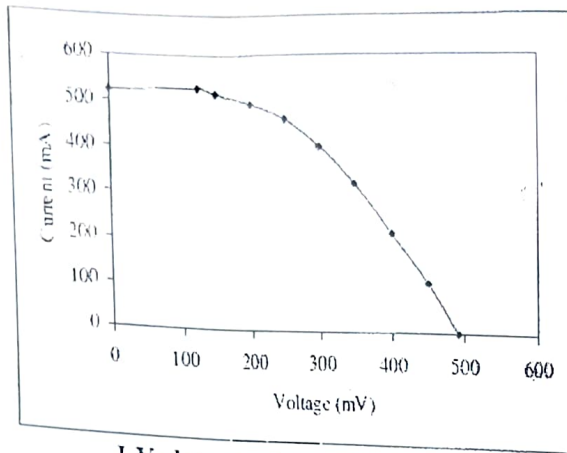
Apparatus Used

Solar cell experimental setup (see inner cover page) consisting of digital dc ammeter 0-2A, digital dc millivoltmeter 0-2000mV, Light source and light arrangements to vary the distance between the source and solar cell, Solar cell 10cm dia circular, ac power supply 12V/5A and a rheostat 0-10 W/3A.

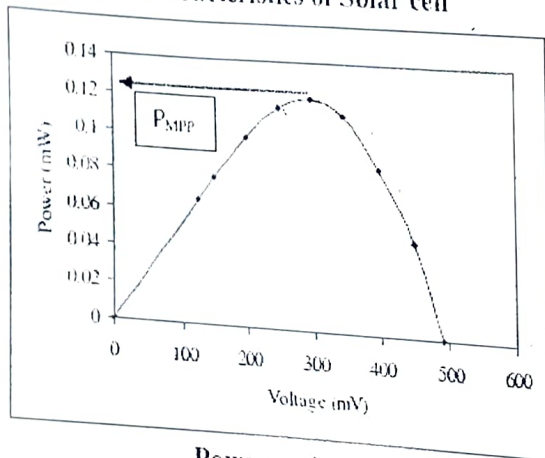
Experimental Procedure

1. The circuit connections are made as shown in Figure-5. The solar cell illuminator power supply is connected to the bulb and bulb is switched on. The cooling fan is also switched on.
2. The solar cell is placed under the halogen lamp at distance of 30cm from lamp.
3. The rheostat is shorted ($R=0$) and the short circuited current is noted. This is current is I_{sc} .
4. Both the terminals of the rheostat are disconnected ($R=\infty$) from the ammeter voltmeter junction and the open circuit voltage is noted in the millivoltmeter. This is V_{oc}
5. The rheostat is reconnected and it is adjusted such that voltage is around 100mV. The corresponding current is noted.
6. Trial is repeated by setting voltage to 150mV, 200mV etc up to maximum of 450mV by varying rheostat. The corresponding current noted is recorded in Table-1. Power is calculated in each trial and maximum power point for which output power is the maximum is noted
7. Experiment is repeated by keeping the solar cell at 27cm from the light source. The corresponding readings obtained are tabulated.
8. A curve is drawn taking readings from Table 1 with voltage along X-axis and current along Y-axis.
9. Another graph of power versus voltage is drawn to see the power variation. The curve resembles the maximum power transfer theorem.
10. The experiment is repeated for $d=23, 19$ and 15 cm from the source, and the readings obtained are tabulated.

Model Graphs



I-V characteristics of Solar cell



Power variation

Observations:-

Resistance (Ω)	Voltage (mV)	Current (mA)	Power (W)

Result

- The IV graph for a solar cell has been plotted
- Power variation with voltage is studied, maximum power point is = _____

Experiment No: 7**Diffraction Grating - Determination of wavelength of Diode laser****Aim**

To determine the wavelength of the Diode laser light using diffraction grating.

Apparatus used

Diode laser, Source, Grating, A screen on which graph paper is pasted optic bench.

Theory

A grating consist of alternate transparent and opaque spaces. If a is the width of the transparent space and b width of opaque spaces, then $a + b$ is called the Grating element. The number of lines per unit length, $N = 1/(a+b)$

If θ_m is the angle of diffraction corresponding to the direction of m^{th} principal maximum, then

$$\sin \theta_m = Nm \lambda$$

Where λ is the wave length of the laser beam, m is the order.

$$\text{Also } \sin \theta_m = X_m / \sqrt{X_m^2 + D^2}$$

Where X_m is the distance from the central spot to the m^{th} order spot, D is the distance between the grating and screen.

$$\text{From the above equations } \lambda = \sin \theta_m / Nm = 1/Nm * X_m / \sqrt{X_m^2 + D^2}$$

If N is known, λ can be calculated.

Procedure

Diode laser source is placed horizontally on an optic grating is mounted normal to the diode source at a distance of about 20 cm. A vertical screen on which centimetre graph paper is pasted is placed at a large distance D from the grating (about m) as shown in figure. Diode source is switched on. Diode laser beam is incident normally on the grating. Diffraction of laser beam takes place and diffraction pattern is obtained on the screen. The zeroth order (centre) dot has maximum intensity and intensity gradually decreases with higher orders. The distance between the centre dot and higher order dots on the both sides are measured and mean x is determined. The distance between the grating and screen D is also measured. The number of lines per meter N is also noted. It may also be specified in the grating data. Hence wave length of diode laser can be determined.

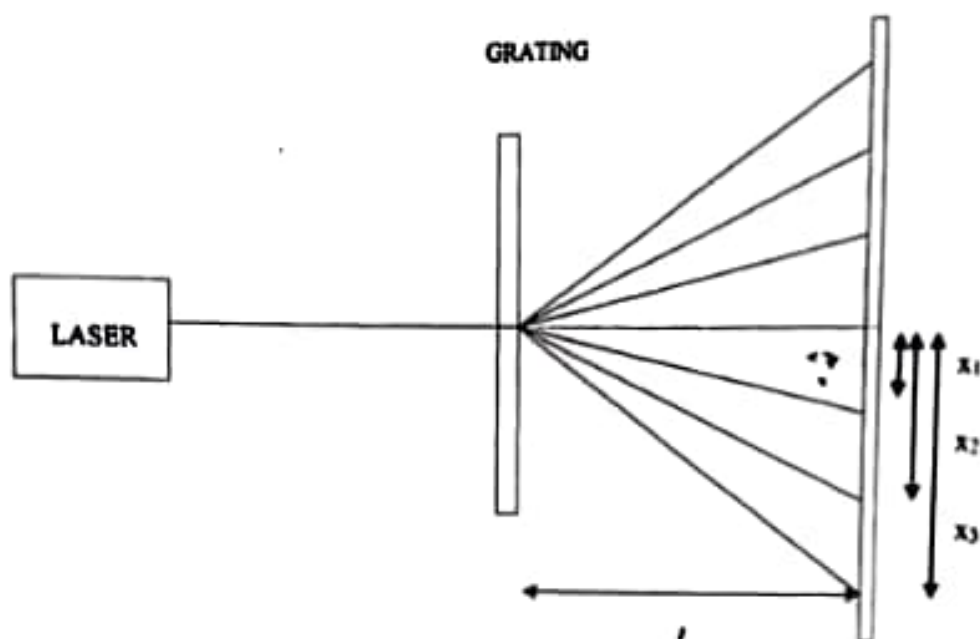


Fig. 1. Experimental Setup

To find λ

S. No.	Order n	D cm	x_m cm			$\sin \theta_m = x_m / \sqrt{x_m^2 + D^2}$	$\lambda = \sin \theta_m / Nm$
			LHS	RHS	Mean		
1	1						
2	2						
3	1						
4	2						
5	1						
6	2						

Mean value of $\lambda = m$.

Calculation

Distance of the spot from the central maximum (x_m) =m

Distance of the screen from grating (D) =m

Number of lines per metre in the grating (N) =lines/m

Wavelength of laser light, $\lambda = \sin\theta / m N = \dots\dots\dots\text{nm}$

Result

Wavelength of Diode laser light, $\lambda = \dots\dots\dots\text{nm}$